Knowledge representation for efficient learning and communication

Jan Bielecki, Warsaw Python Pizza, 20260509

Learning platforms for computer science

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BRILLIANT

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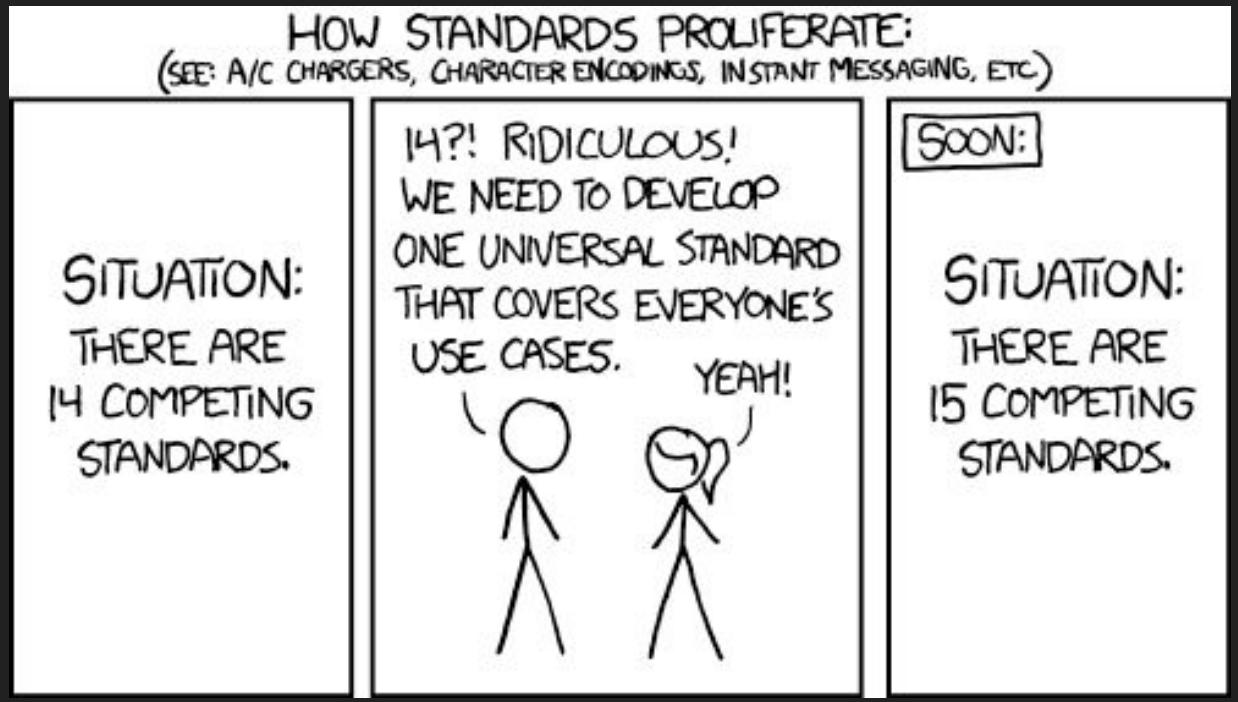
LeetCode

Learning platforms for computer science

Let's create the universal platform!



Gamification of a learning process

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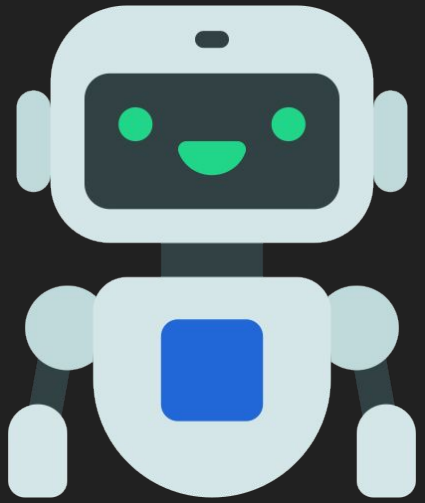
✓ Fun to play

Gamification of a learning process

- ✓ Fun to play
- ✓ Seems useful

Creating a learning content using LLMs



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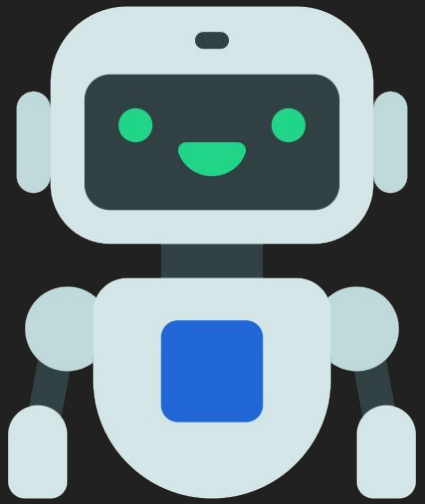
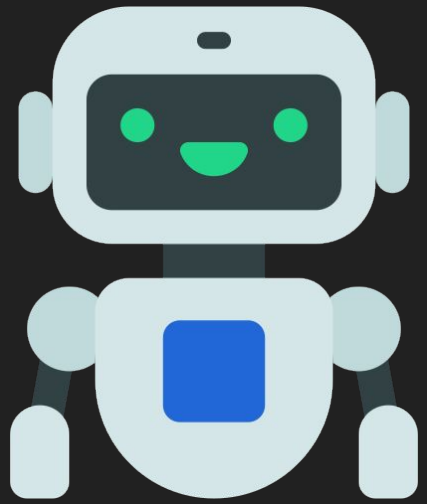


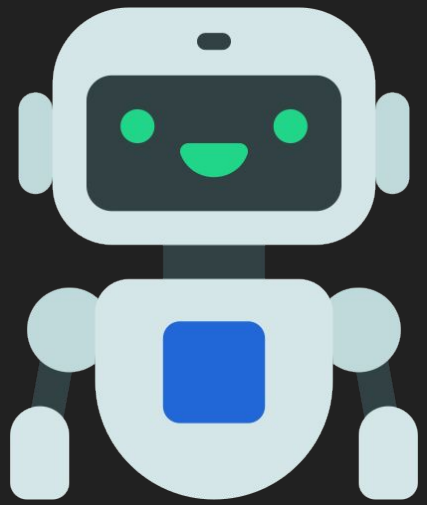
Looks good on a first sign

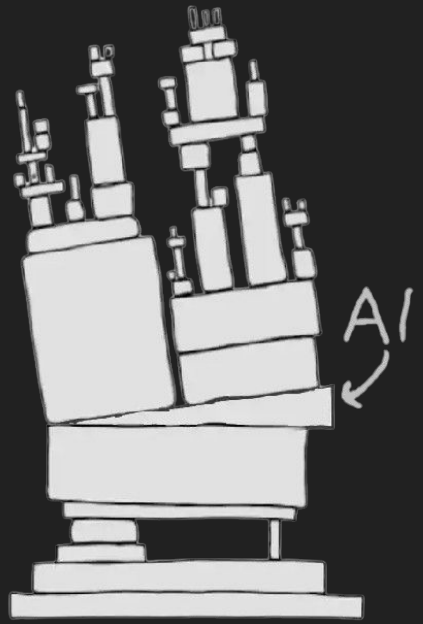
Creating a learning content using LLMs



- ✓ Looks good on a first sign
- ✓ Fast

Creating a learning content using LLMs





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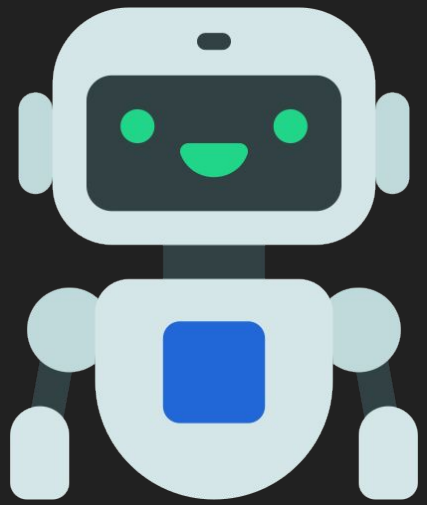
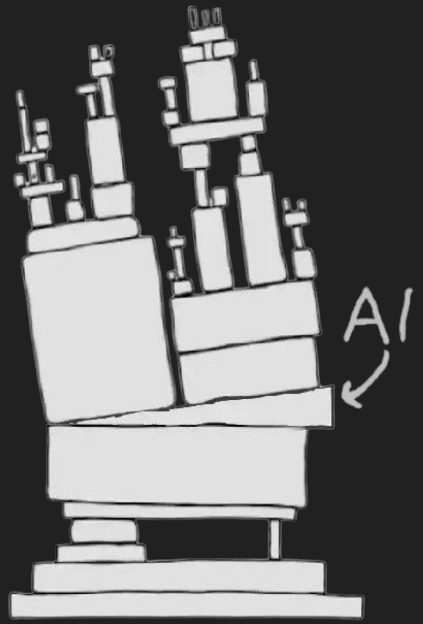
- ✓ Fast
- ✗ Questionable quality

Creating a learning content using LLMs





- ✓ Looks good on a first sign
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- ✗ Questionable quality
- ✗ Luck of a architecture/structure

Creating a learning content architecture/structure

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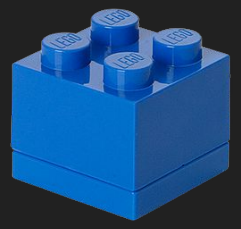
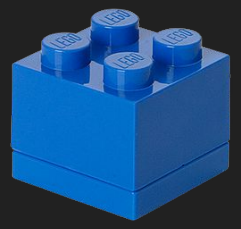


Minimal building block (a basic unit of knowledge)

Creating a learning content architecture/structure



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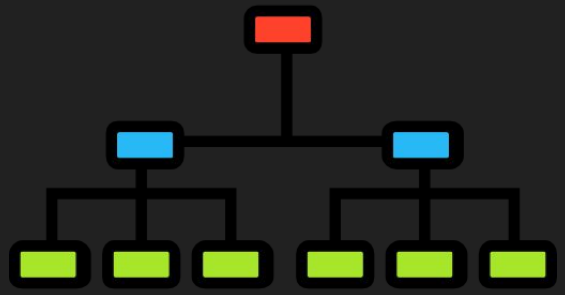
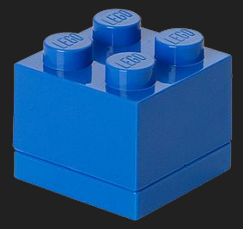


Advancement hierarchy

Creating a learning content architecture/structure



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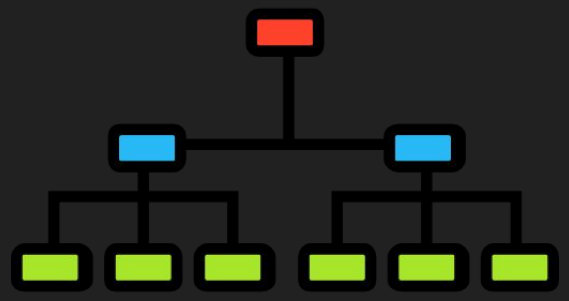
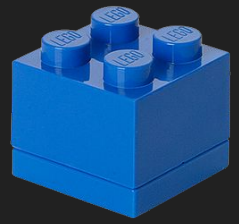


Advancement hierarchy

Modularity

Basic unit of knowledge



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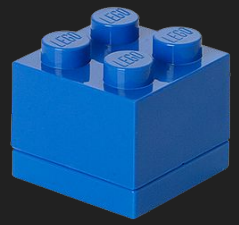
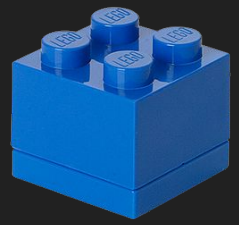


Definition (concept/term)

Basic unit of knowledge



Definition (concept/term)

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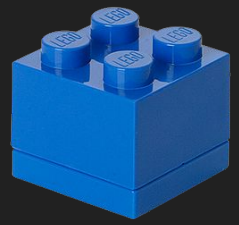

Definition (concept/term)

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- content (can include other definitions)

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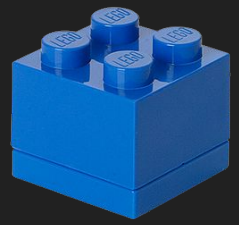
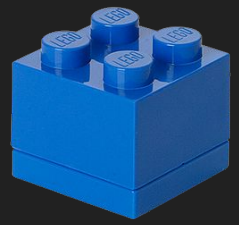


Definition (concept/term)

- name
- content (can include other definitions)
- field (a unique key = name + field)

Basic unit of knowledge



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class _FiniteSet(Definition):
    def _get_content(self) -> str:
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Finite set is a {SET.key.get_reference(phrase="set")} that has a finite number of elements.
Informally, a finite set is a set which one could in principle count and finish counting.
For example, {{2,4,6,8,10}}  is a finite set with five elements.
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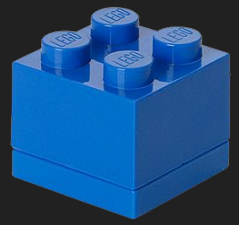

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field
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      analysis
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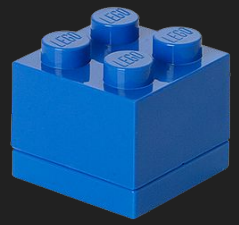

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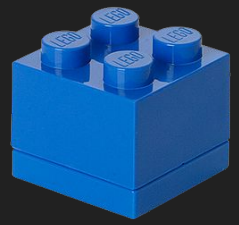

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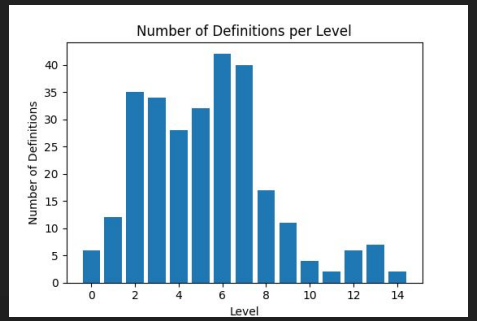


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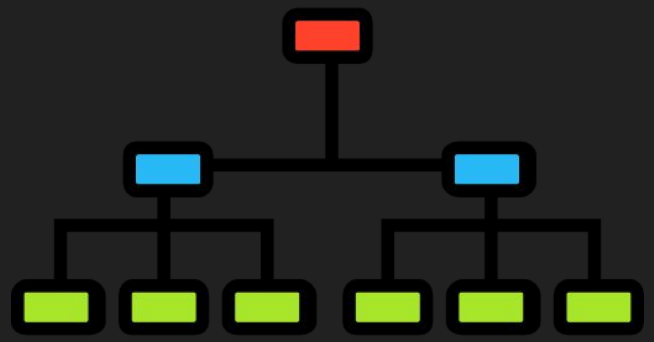
Advancement hierarchy



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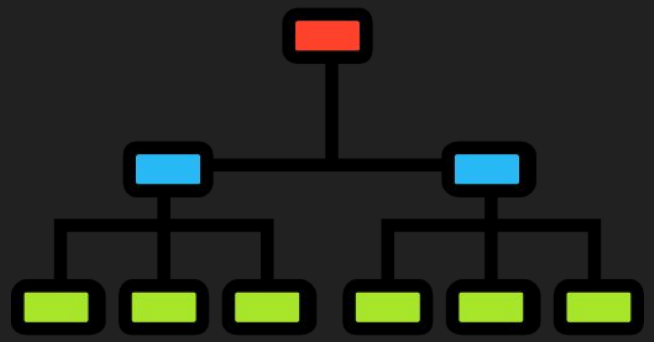
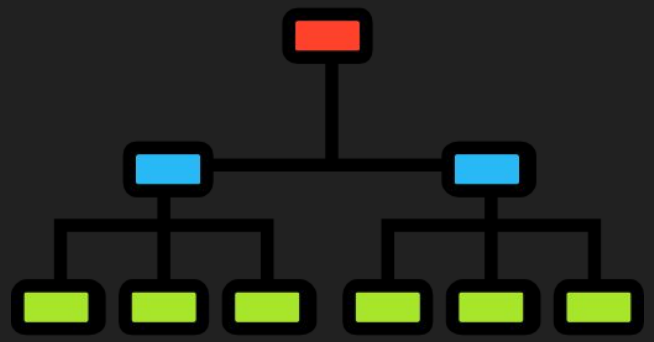


Definitions together with references (dependencies) form a graph.

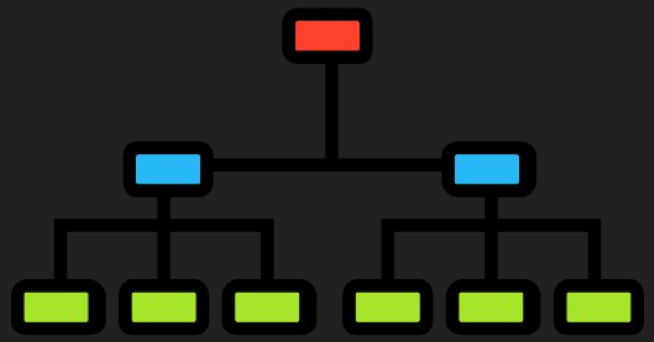
Advancement hierarchy



Definitions together with references (dependencies) form a graph.

1) Definition -> node

Advancement hierarchy



Definitions together with references (dependencies) form a graph.

1) Definition -> node
2) Dependency -> directed edge

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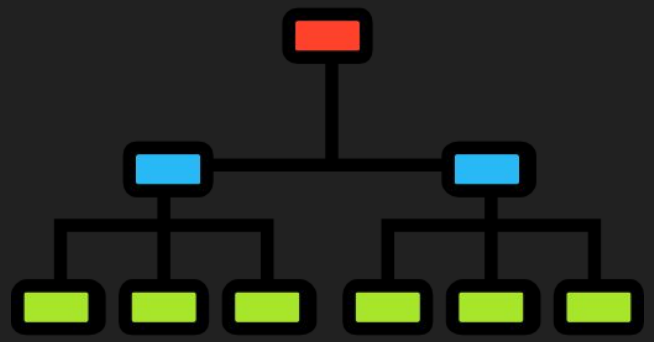

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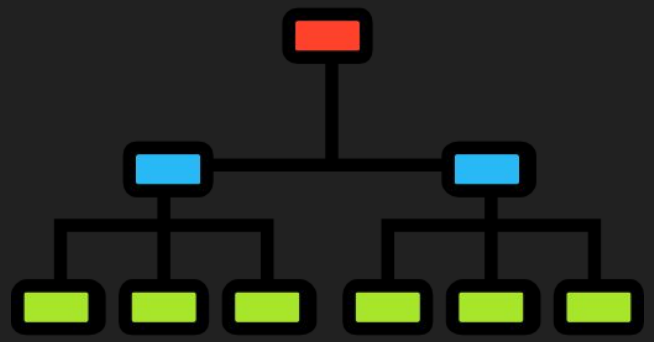
3) Rule -> no cycles

Advancement hierarchy



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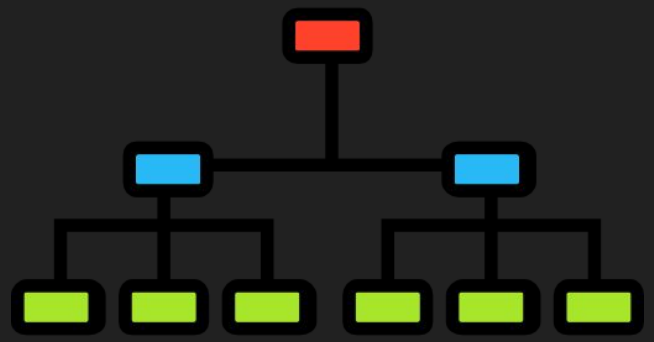
Directed Acyclic Graph (DAG)

Modularity



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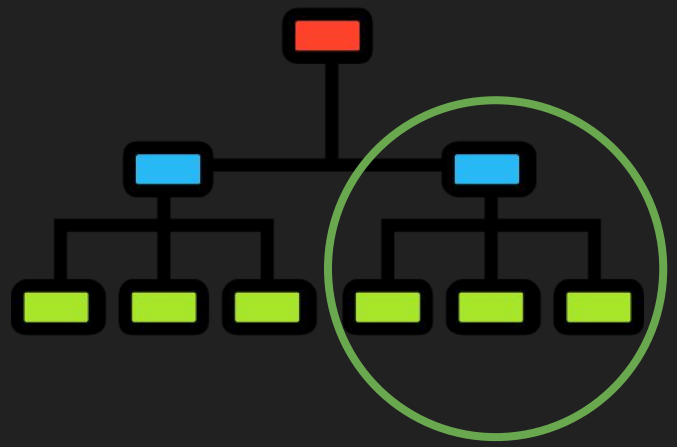

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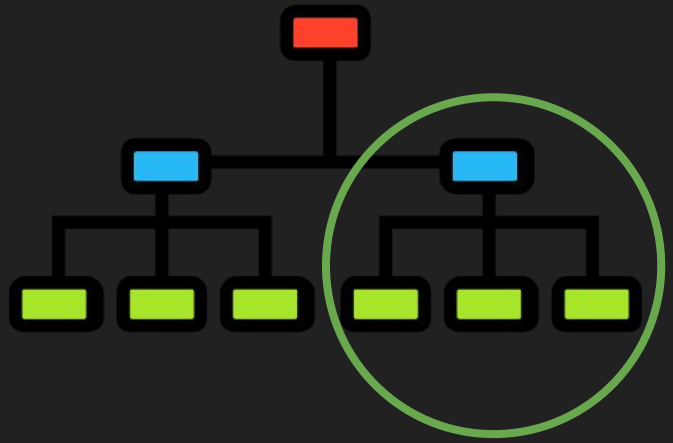

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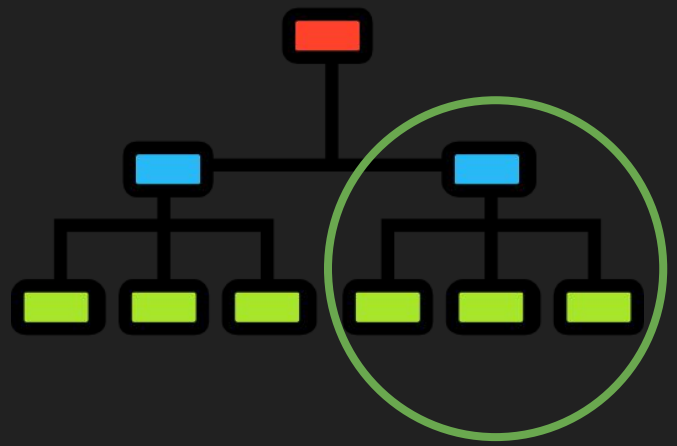
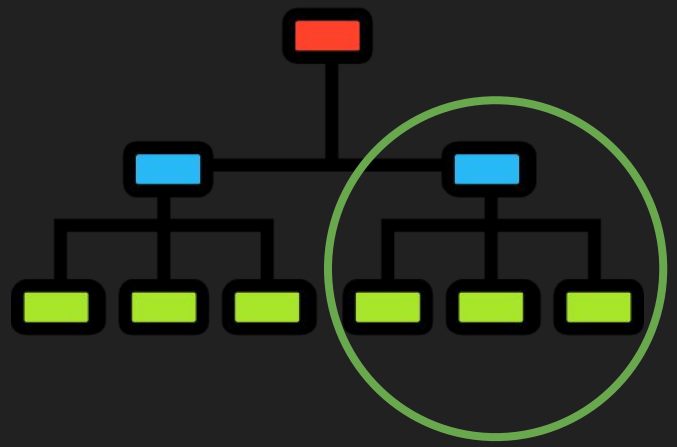


Data Structures & Algorithms

Modularity



Data Structures & Algorithms

Computer hardware

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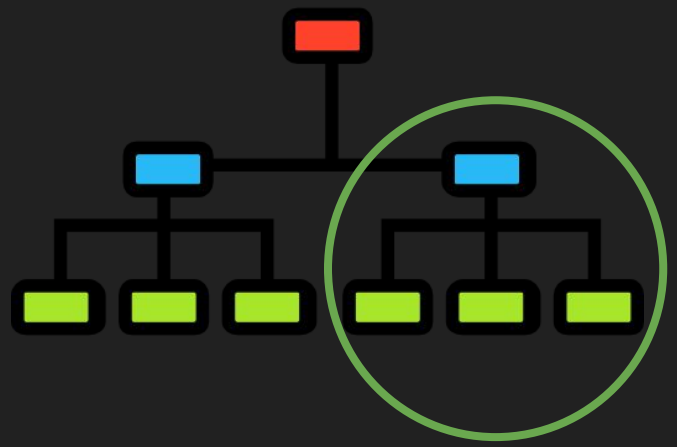


Data Structures & Algorithms

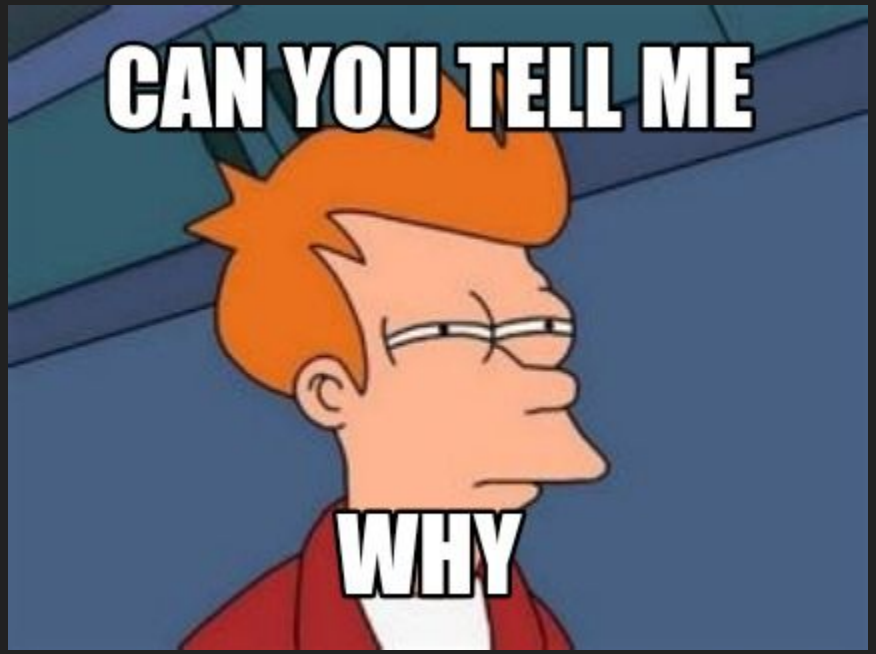
Computer hardware

Databases

POC - Data Structures and Algorithms

k4liber.github.io/definit-dsa

What is the value?



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1) Breaking down (decompose) complex topics into small “knowledge pills”.

2) DAG structure suggests “where to start”.

3) A living and high-quality glossary for any field of knowledge used as reference or ML training data.

The biggest challenges

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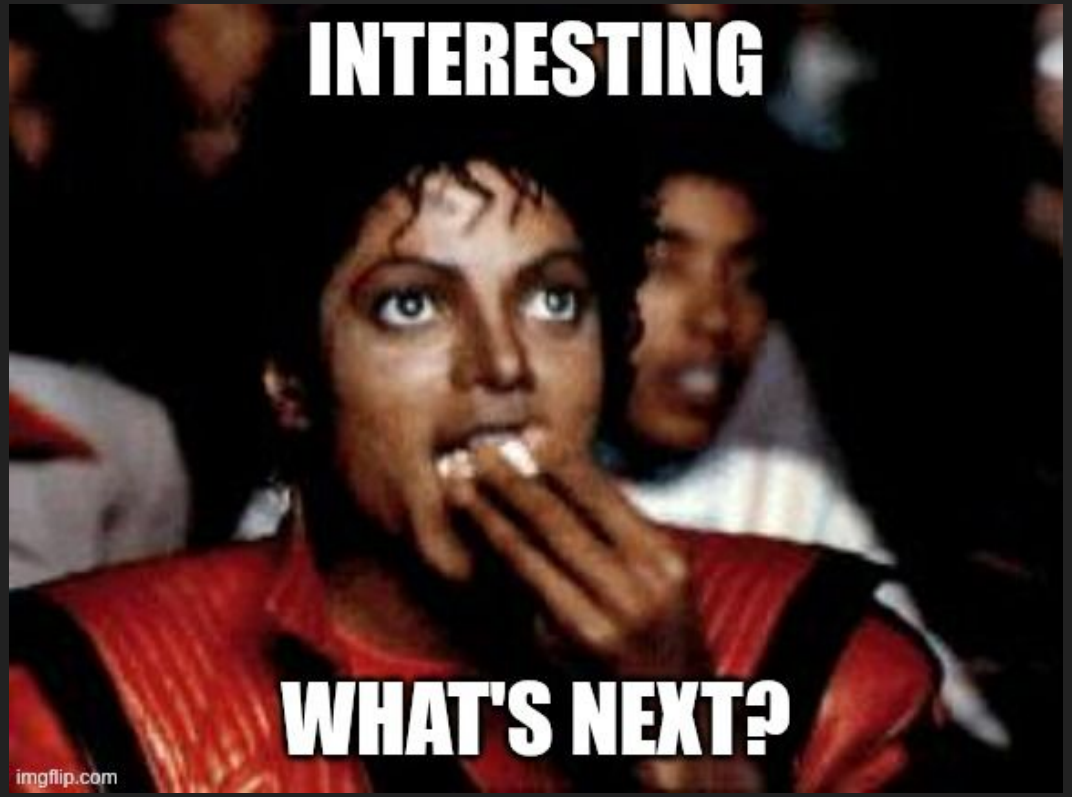
2) "In theory, there is no difference between theory and practice. But in practice, there is."~Benjamin Brewster

What’s next?



Facing the challenges

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Facing the challenges

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<u>Creating next modules to verify the assumption.</u>

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Facing the challenges

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Creating next modules to verify the assumption.

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Vibe-coding gamification aspects: quests, badges, ranks.

"You can outsource your thinking, but you cannot outsource your understanding."

github.com/K4liber/definit